CyberGuard Sentinel v2.2.3

Administrator Guide

Installation, Configuration & Operations

CONFIDENTIAL — For authorized administrators only

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1. Prerequisites

1.1 System Requirements

Component	Requirement
OS	Ubuntu 22.04/24.04 LTS or Debian 12/13
RAM	2 GB minimum, 4 GB recommended
Disk	20 GB minimum
CPU	2 cores minimum
Network	At least one interface on the monitored subnet
Python	3.12 or higher (Ubuntu 24.04 ships 3.12, latest stable is 3.14)
Access	Root or sudo privileges

1.2 Network Position

The Sentinel server must be on the same network segment as the machines it monitors. It uses ARP sweeps for discovery — ARP does not cross routers.

> ⚠️ *If your network has multiple VLANs, you need either one Sentinel per VLAN or a mirrored port (SPAN) on your switch.*

1.3 Before You Start

- Ensure you have SSH access to the server with a key (not just password)
- Know the subnet(s) to monitor (e.g. 192.168.1.0/24)
- Have SMTP credentials ready if you want email notifications
- Decide on a defense mode: Confirmation (safest) or Immediate

2. Installation via SSH

2.1 Transfer the Package

From your local machine, copy the .deb package to the server:

```
scp cyberguard-sentinel_2.2.3-1_all.deb user@server:/tmp/
```

Then connect via SSH:

```
ssh user@server
```

2.2 Install

```
sudo apt update
sudo dpkg -i /tmp/cyberguard-sentinel_2.2.3-1_all.deb
sudo apt install -f    # Install missing dependencies if any
```

The installation runs 7 steps:

- Step 1: System dependencies (nmap, whois, traceroute)
- Step 2: Python virtual environment creation
- Step 3: Python library installation (scapy, flask, bcrypt...)
- Step 4: Interactive setup wizard (see section 3)
- Step 5: TLS certificate generation + SSH hardening + OS hardening
- Step 6: Systemd service creation
- Step 7: Atomic commit (all-or-nothing)

> ⚠️ *If your SSH connection drops during installation, the system rolls back automatically. No partial install is possible. Simply reconnect and run the dpkg command again.*

2.3 What the Installer Does Automatically

- Generates a self-signed TLS certificate for the web dashboard
- Hardens SSH: disables password authentication if SSH keys are detected
- Hardens the OS: sysctl network protection, disables USB storage, enables audit rules
- Creates a systemd service with sandboxing (ProtectHome, PrivateTmp, etc.)

3. Setup Wizard

The wizard runs automatically on first install. To re-run it later:

```
sudo cyberguard setup
```

It has 9 sections:

3.1 Network

The wizard auto-detects your subnet and interface. Verify and adjust:

- Subnets to monitor (CIDR format, e.g. 192.168.1.0/24)
- Capture interface (e.g. eth0, ens33)
- IPs to exclude from scanning (your server IP is excluded automatically)
- Nmap: keep or remove (scapy-only mode if removed)

3.2 Suricata (Optional)

If you have a Suricata IDS, Sentinel can ingest its alerts. This is 100% optional — Sentinel has its own detection engine.

3.3 Email Notifications

Configure SMTP for incident emails:

- SMTP server, port, TLS, credentials
- Admin email addresses (receive all alerts)
- User directory: associate each workstation IP/MAC with a user email
- Country for legal information (IE, FR, US)

3.4 Client Agent

Optional zero-privilege agent for workstation popups and local forensic collection. See section 7.

3.5 Active Defense

Choose the defense mode:

- Confirmation (default): admin must approve before Sentinel acts
- Immediate: Sentinel acts instantly, informs admin with rollback option

Also configure: auto-block thresholds, whitelist IPs, block duration.

3.6 – 3.9

- Reconnaissance: optional AbuseIPDB and VirusTotal API keys
- Netgate: optional pfSense/OPNsense firewall integration
- Web dashboard: port, username, password
- Data retention: how long to keep alerts, flows, DNS logs

4. Starting the Service

```
sudo systemctl start cyberguard-sentinel
sudo systemctl enable cyberguard-sentinel    # Start on boot
```

Verify it is running:

```
sudo systemctl status cyberguard-sentinel
sudo cyberguard status
```

Check the logs:

```
sudo journalctl -u cyberguard-sentinel -f
```

4.1 What Happens at Startup

The daemon performs these checks in order:

- Verifies code integrity (SHA-256 manifest)
- Enables SQLite WAL mode (concurrent reads/writes)
- Verifies SSH is hardened (password auth disabled if keys exist)
- Verifies OS hardening is in place (sysctl, modules, etc.)
- Starts packet capture (requires root, then drops privileges)
- Starts the web dashboard on HTTPS
- Begins ARP discovery and threat detection

5. Web Dashboard

Access the dashboard at:

```
https://<SERVER_IP>:8443
```

> ⚠️ *The certificate is self-signed. Your browser will show a warning — this is normal. Add an exception to continue.*

5.1 Authentication

- Username and password set during setup wizard
- Optional TOTP 2FA (Google Authenticator compatible)
- Account locks after 5 failed attempts (15 min lockout)
- Session expires after 30 minutes of inactivity

5.2 Dashboard Features

- Real-time alert feed via WebSocket
- Host inventory with risk scores and identity confidence
- Active defense rules (iptables)
- Incident management with approval/rejection
- Approval PIN display (for email link verification)
- Snapshot list and rollback
- Kill chain detection status
- Server health monitoring

5.3 Key API Endpoints

Endpoint	Description
GET /api/overview	System overview and stats
GET /api/alerts	Alert list
GET /api/hosts	Host inventory
GET /api/incidents	Incident list
GET /api/incidents/<id>/pin	Approval PIN (dashboard only)
GET /api/snapshots	Available rollback snapshots
POST /api/snapshots/rollback	Restore a snapshot
GET /api/resilience	Server health + degraded mode
GET /api/integrity	Code integrity check
GET /api/ssh/verify	SSH hardening status
GET /api/os/verify	OS hardening status

6. TUI Console (SSH Administration)

For servers without a browser, use the SSH console:

```
sudo cyberguard console
```

Menu options:

- 1. System status — CPU, RAM, disk, service state
- 2. Recent alerts — last 50 alerts with severity
- 3. Network inventory — all discovered hosts with risk scores
- 4. Incidents — active and resolved incidents
- 5. Active defense — iptables rules + ROLLBACK option
- 6. Modify configuration — re-run wizard or edit YAML
- 7. User directory — add/edit/remove users, scan network
- 8. Test email — send a test to verify SMTP
- 9. View logs — daemon log, defense audit, forensic files
- 10. Restart service

6.1 Performing a Rollback

From the TUI console: option 5 (Active defense) → option 1 (Rollback)

The system shows available snapshots. Select one to restore the firewall state from before that incident's defense actions.

7. Client Agent Deployment

7.1 Overview

The client agent is optional. Without it, everything works via email. With it, users get instant popup alerts and local forensic collection.

7.2 Architecture

- Sentinel NEVER connects to workstations
- The agent polls Sentinel for notifications (outbound only)
- Adaptive polling: 60s when idle, 5s when a notification is pending
- HMAC-SHA256 mutual authentication (shared secret)
- Popup messages are predefined in the agent — server sends type codes only
- Forensic reports save to Desktop only — never sent anywhere

7.3 Deployment

Copy cyberguard-agent.py to each workstation. The shared secret is shown during setup or in config.yaml:

```
python3 cyberguard-agent.py --server https://SENTINEL_IP:8443 --secret
YOUR_SECRET
```

No admin privileges needed. The agent runs as a normal user.

> ⚠️ *If a workstation is compromised, only its shared secret is exposed. Consider using per-workstation secrets in high-security environments.*

8. Defense Modes

8.1 Confirmation Mode (Default)

Safest option. The workflow is:

- 1. Sentinel detects a threat
- 2. Email sent to admin with Approve/Reject buttons
- 3. Admin clicks Approve → enters PIN from dashboard
- 4. Defense actions execute (block IP, quarantine, sinkhole...)
- 5. User notified (popup if agent, email otherwise)
- 6. Report sent to admin with rollback option

8.2 Immediate Mode

For environments that need instant response:

- 1. Sentinel detects a threat
- 2. Defense actions execute IMMEDIATELY
- 3. User notified in parallel
- 4. Admin receives email AFTER the fact with full details + rollback option

> ⚠️ *In immediate mode, a false positive will block a legitimate IP instantly. Use the rollback feature to undo.*

8.3 Switching Modes

```
sudo cyberguard console    # Option 6 > Edit YAML
```

Change defense.mode to "confirmation" or "immediate" in config.yaml, then restart.

9. Incident Workflow

9.1 Confirmation Mode Flow

Status	What Happens	Admin Action
DETECTED	Threat identified, actions planned	None yet
AWAITING_ADMIN	Email sent with Approve/Reject	Click link + enter PIN
APPROVED	Defense + notification launched	Wait for report
MITIGATING	Recon, block, forensic, sinkhole	Monitor dashboard
RESOLVED	Report emailed with rollback option	Review, rollback if needed

9.2 Approval PIN

When an incident is created, a 6-digit PIN appears in the dashboard. The approval email link requires this PIN. Even if the email is intercepted, the attacker cannot approve without the PIN.

9.3 What Defense Actions Do

- Block IP: adds iptables DROP rule for the attacker IP
- Rate limit: limits packets/second from an IP
- Quarantine: isolates an internal compromised host
- DNS sinkhole: redirects a malicious domain to 127.0.0.1
- If Netgate is configured: pushes blocks to pfSense/OPNsense firewall too

10. Rollback & Snapshots

Before every defense action, Sentinel saves a complete snapshot of:

- All iptables rules (iptables-save)
- All nftables rules
- /etc/hosts (sinkhole entries)
- CyberGuard-specific iptables chain

10.1 How to Rollback

Three methods:

- TUI Console: option 5 > Rollback > select snapshot
- Web API: POST /api/snapshots/rollback with filepath
- Email link: rollback instructions in every incident report

> ⚠️ *A rollback removes ALL blocks added since the snapshot. If multiple incidents occurred, rolling back to an early snapshot may unblock IPs from later incidents too.*

11. Maintenance

11.1 Automatic Tasks

Task	Frequency	Notes
ARP discovery	Every 5 min	Configurable
Port scan	Every 60 min	Configurable
Self-monitor	Every 60s	CPU, RAM, disk, I/O
Buffer guard	Every 60s	Packet drop detection
Firewall verify	Every 2 min	Tamper detection
SSH verify	Every 30 min	Password auth check
OS hardening verify	Every hour	Sysctl drift detection
Data cleanup	Daily 3:00 AM	Per retention settings
Backup	Daily 4:00 AM	Low-priority I/O
Weekly report	Monday 8:00 AM	Emailed to admins

11.2 Manual Tasks

- Review weekly reports for trends
- Update detection rules: edit /etc/cyberguard/rules.yaml (hot-reloaded)
- Update threat intel API keys if expired
- Check disk space on /var/log/cyberguard/
- Review false positives and mark them via API or TUI

11.3 Reconfiguration

```
sudo cyberguard setup       # Full wizard
sudo cyberguard console     # Option 6 for quick edits
sudo systemctl restart cyberguard-sentinel   # Apply changes
```

12. Troubleshooting

12.1 Service Won’t Start

```
sudo journalctl -u cyberguard-sentinel -n 50
```

- "Permission denied" → must run as root (raw sockets)
- "Address already in use" → port 8443 occupied, change web.port
- "Module not found" → run: sudo /opt/cyberguard/venv/bin/pip install -r /opt/cyberguard/requirements.txt

12.2 No Hosts Discovered

- Verify Sentinel is on the same subnet (ARP doesn’t cross routers)
- Check the capture interface in config.yaml
- Try manual scan: sudo cyberguard scan

12.3 Emails Not Sending

```
sudo cyberguard console    # Option 8: Test email
```

- Check SMTP credentials in config.yaml
- Verify firewall allows outbound port 587/465

12.4 Dashboard Unreachable

- Verify service is running: systemctl status cyberguard-sentinel
- Check port: ss -tlnp | grep 8443
- Check firewall: sudo iptables -L INPUT -n | grep 8443

12.5 Degraded Mode Activated

This means the server is overloaded. Non-essential tasks are suspended.

- Check what’s consuming resources: top, iotop
- If backup is running: it uses low I/O priority, wait for completion
- Manual exit: POST /api/resilience/degraded {"action": "exit"}

13. Security Architecture

13.1 Zero-Privilege Design

- Sentinel NEVER connects to workstations (no SSH, no WinRM)
- No credentials for client machines stored anywhere
- Client agent polls Sentinel (outbound only)
- Forensic reports stay on user's Desktop (never uploaded)

13.2 Defense in Depth

Layer	Protection
Network	HTTPS only, CORS restricted, rate limiting
Authentication	bcrypt + TOTP 2FA + login lockout + session timeout
Authorization	All 46 API endpoints authenticated
Approval	Token + PIN (second factor, dashboard only)
Agent	HMAC-SHA256 + anti-replay + predefined messages
Data	SQLite WAL, encrypted backups, log sanitization
Audit	Hash-chained tamper-proof log, admin action audit
OS	Sysctl hardening, AppArmor, auditd, SSH hardening
Integrity	SHA-256 manifest checked at startup
Resilience	Degraded mode, buffer guard, safe backups

13.3 File Locations

Path	Contents
/opt/cyberguard/	Application code + virtualenv
/etc/cyberguard/config.yaml	Configuration
/etc/cyberguard/tls/	TLS certificate and key
/var/lib/cyberguard/data/	SQLite database + fingerprints
/var/log/cyberguard/	Logs + defense audit
/var/log/cyberguard/forensics/	Forensic reports + complaint PDFs
/var/log/cyberguard/snapshots/	Defense state snapshots
/var/log/cyberguard/backups/	Daily backups

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